

250MW capacity auction at Bhadla in Rajasthan. South Africa's Phelan Energy Group and Avaada Power won contracts to build capacities of 50MW and 100MW, respectively, at the Adani Renewable Energy Park Rajasthan.

Falling solar power prices have prompted the Indian government to focus on renewable energy resources. However, new tender announcements from the government have slowed down, as it is encouraging greater private-sector participation. This has intensified the competition amongst developers. Interestingly, only a few new tenders are slated for the next year, and there is a complete lack of pipeline visibility over that period. Therefore it seems that the current tariff rates are hardly sustainable.

Falling tariffs are a double-edged sword for the solar sector. While these make solar power more attractive for consumers, investors and lenders get jittery. These tariffs are also creating uncertainty amongst policymakers, while causing new risks for older projects auctioned at tariffs that are two to three times higher.

The sharp reduction of tariffs may commoditise the Indian solar market rapidly and profit margins are bound to squeeze across the value chain. Therefore solar players need to innovate new technologies and business strategies to survive in this ultra-low-tariff scenario.

Is domestic solar equipment manufacturing a distant dream?

Being a lucrative solar power market, India attracts both global and local investors. However, the country is far from attaining self-reliance in solar module manufacturing. According to Ministry of Commerce data, India imported 5.7GW or about 89 per cent (84 per cent from China alone) of its total solar module requirements in FY 2016-17 (Fig. 4). The total value of these imports

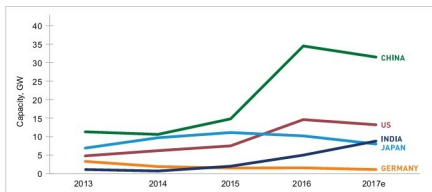


Fig. 1: Solar capacity addition in the leading countries (Source: BTI, GTM Research)

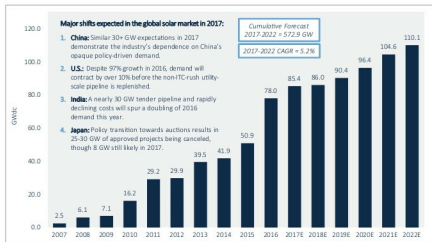


Fig. 2: Global PV demand 2007-22 (Source: GTM Research)

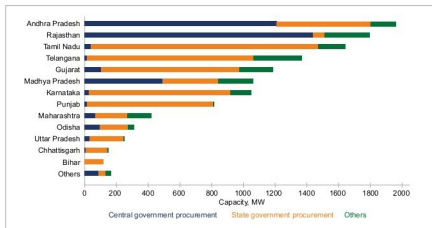


Fig. 3: State-wise commissioned solar capacity as of March 31, 2017 (Source: BTI)

is estimated at US\$3 billion, equivalent to 2.8 per cent of the country's total merchandise trade deficit.

According to a BTI estimate, India's total module manufacturing capacity is 5.286GW. However, most of this capacity is obsolete,

sub-scale and uncompetitive. Total domestic production in 2016 was about 1.33GW. Adani is the biggest module manufacturer with 1.2GW capacity, followed by Waaree, Vikram and Emmvee with capacities of 500MW each.